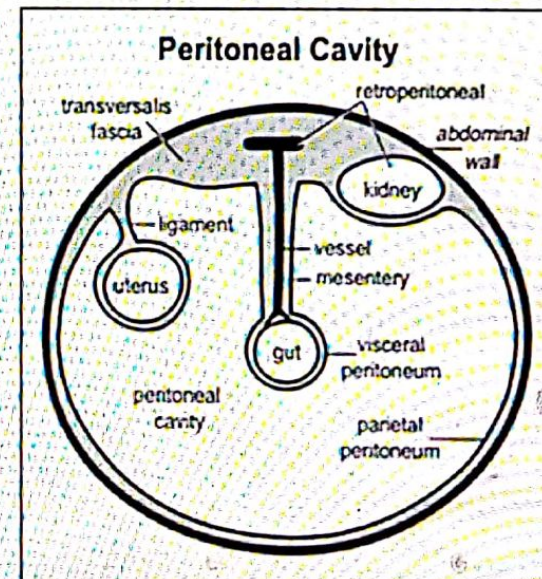
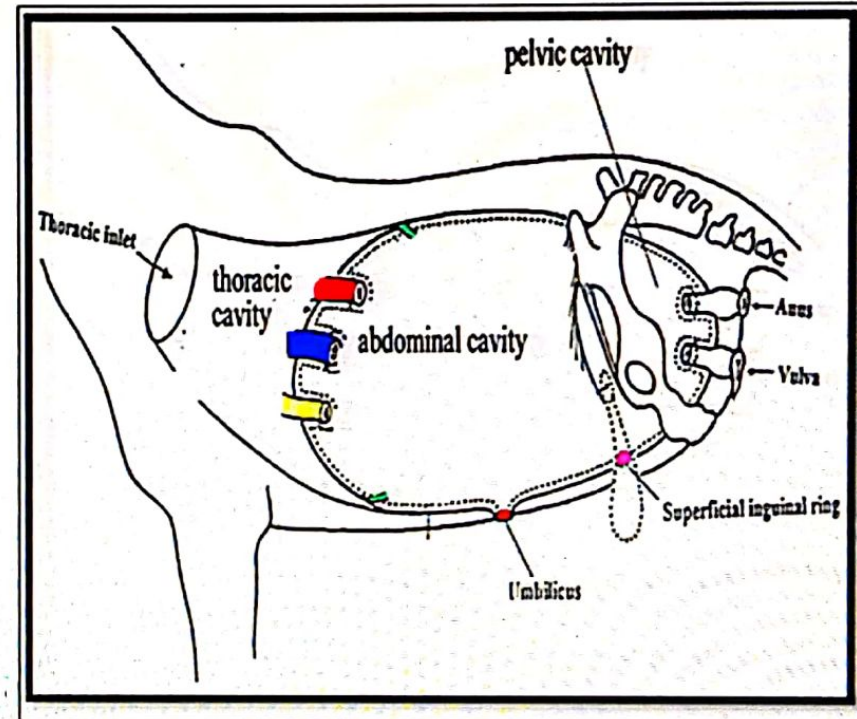


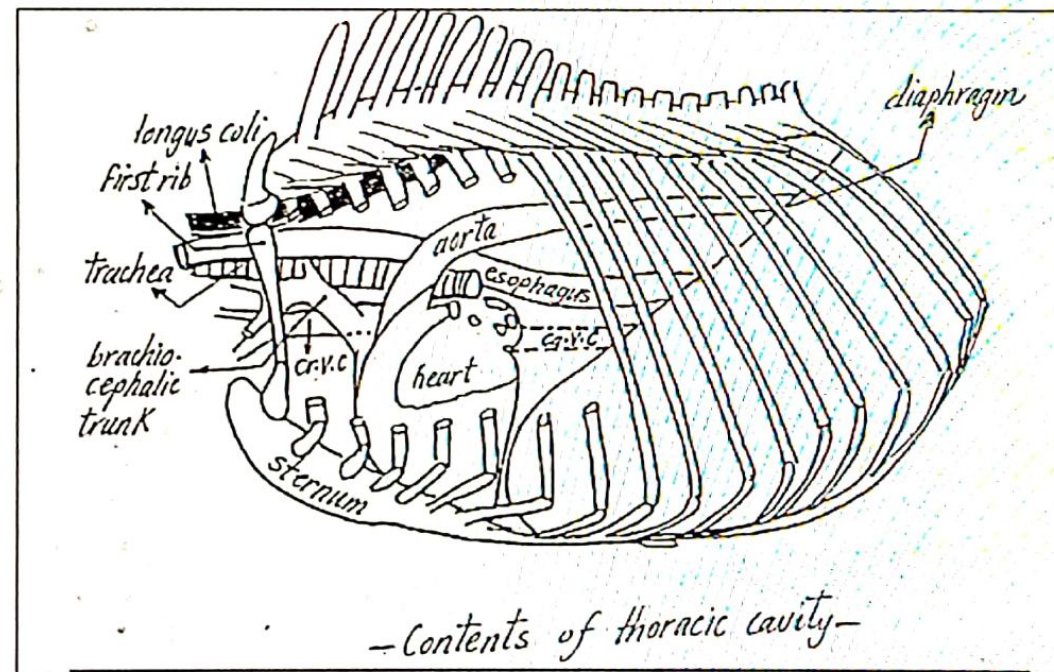
Thoracic cavity

Body Cavities

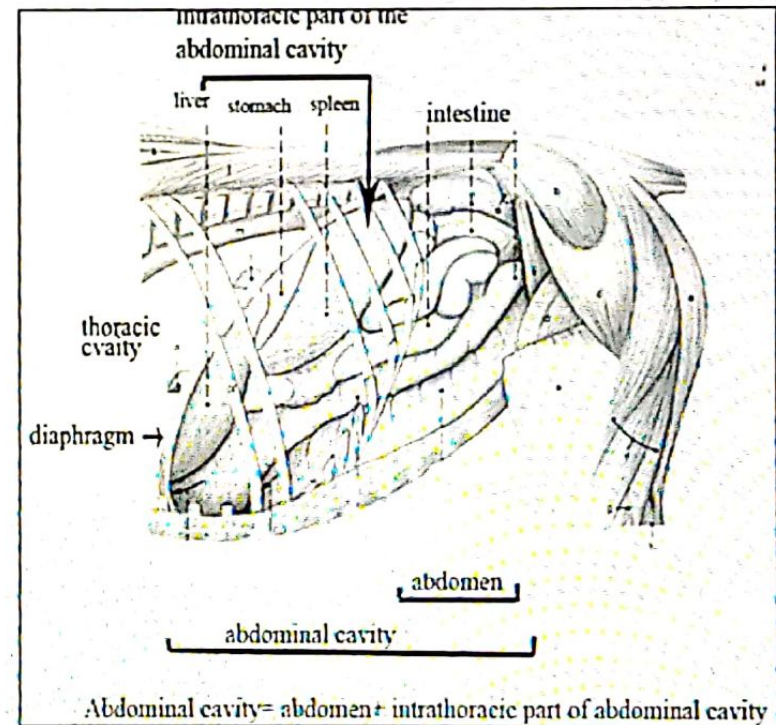
- The mammalian body has three large body cavities in the trunk; **thoracic, abdominal, and pelvic cavities** from cranial to caudal
- The body cavities are lined with serous membrane known as **parietal serous membrane**, pleura or peritoneum, which is reflected to invest their contained organs and known as a **visceral serous membrane**, pleura or peritoneum

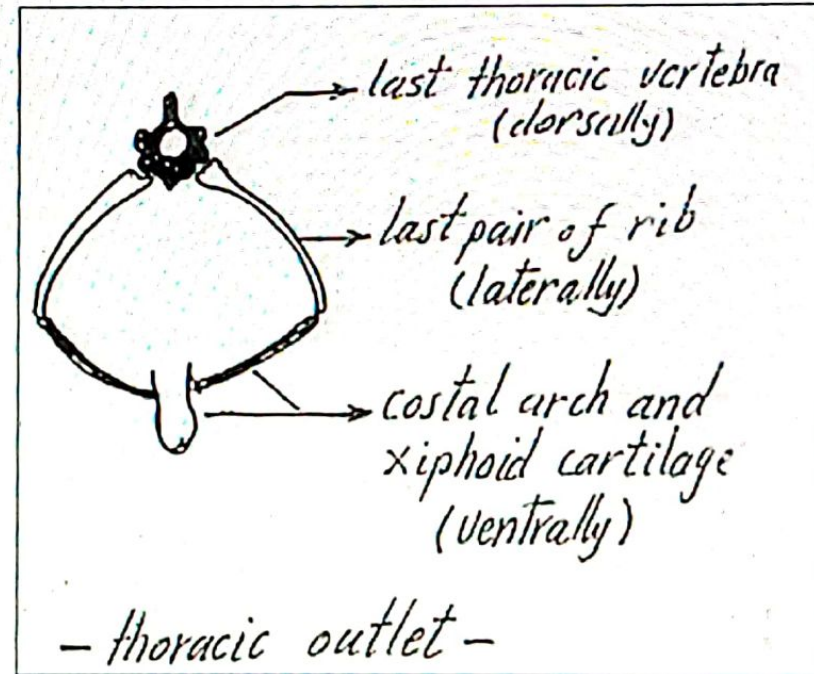
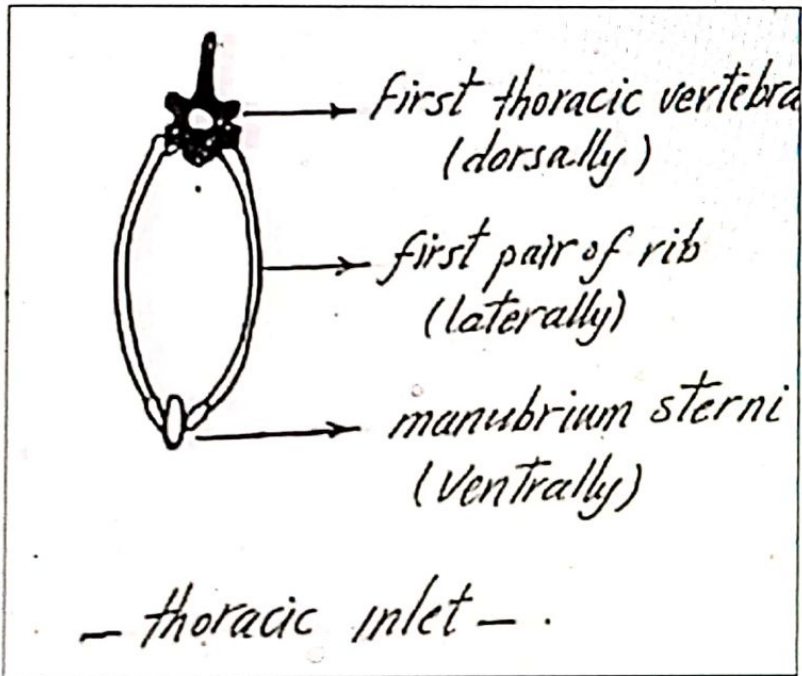


- Thoracic cavity is containing the two lungs in its pleural sac and the heart in its pericardium
- **The shape of thoracic cavity** has the form of truncated cone. Its apex represents the thoracic inlet, which is closed by the trachea, esophagus, longus colli m., vessels and nerves and the fascia in between. While its base represents the thoracic outlet, which is closed by the diaphragm.

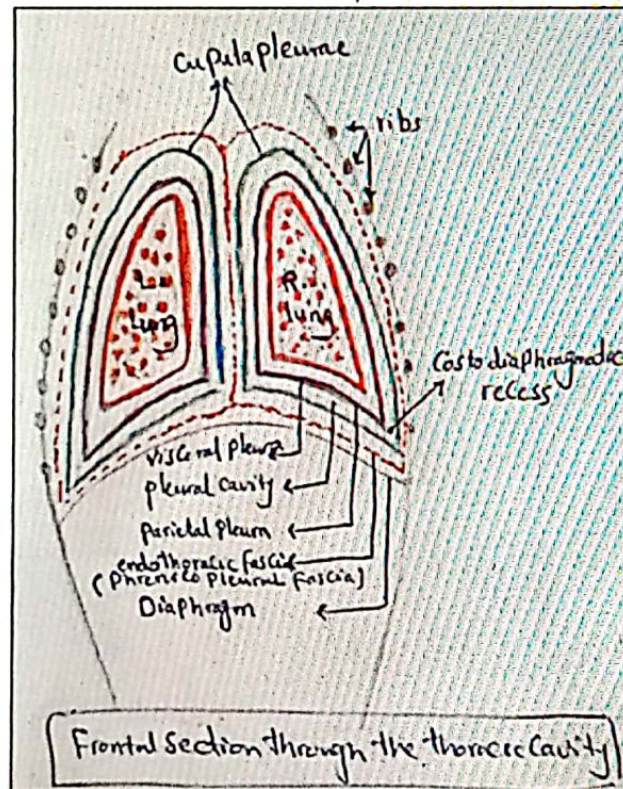


- When the bony thorax and the other components of the wall are present & closed by diaphragm the thoracic cavity is formed.
- The cranial part of the bony thorax contains the thoracic cavity. While, its caudal part contains **an intrathoracic part of the abdominal cavity** contain abdominal organs. So thoracic cavity is smaller than the thorax.



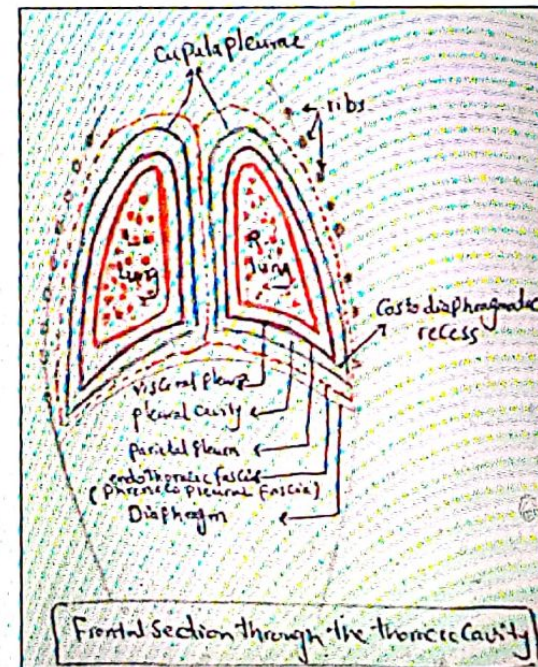
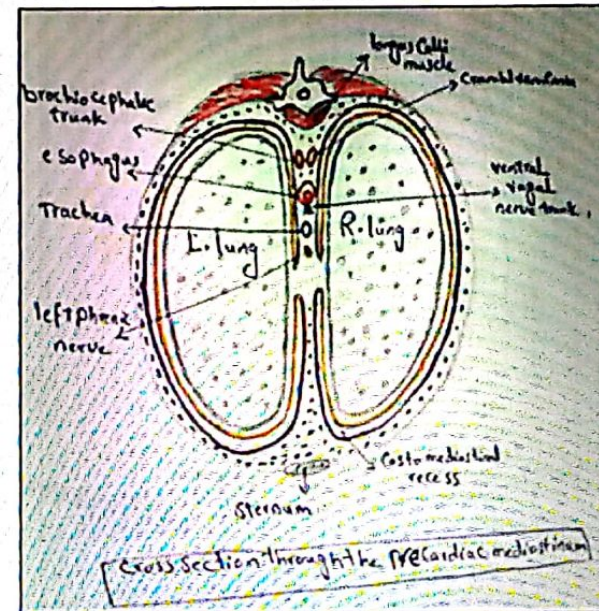


- At the median plane, it is reflected dorsally and ventrally to the thoracic cavity (**mediastinal fascia**) and divided it into two halves to divide the thoracic cavity into two chambers, each chamber is lined with its own serous membrane (**pleura**), which form the pleural sac enclosing its pleural cavity
- **The pleural cavity** is the area between visceral and parietal pleura



Endo-thoracic fascia

- It is loose connective tissue (fibro elastic c.t.) which covers the :
 - ventral surface of the bodies of the thoracic vertebrae
 - inner surface of the ribs with costal cartilage (arch)
 - transverse thoracic muscle
 - thoracic surface of the diaphragm (Phrenico-pleural fascia)
- At the thoracic inlet, it is continuous with the deep cervical fascia.
- So the inflammation can transform from each other



Boundaries of the thoracic cavity

□ Dorsal boundary

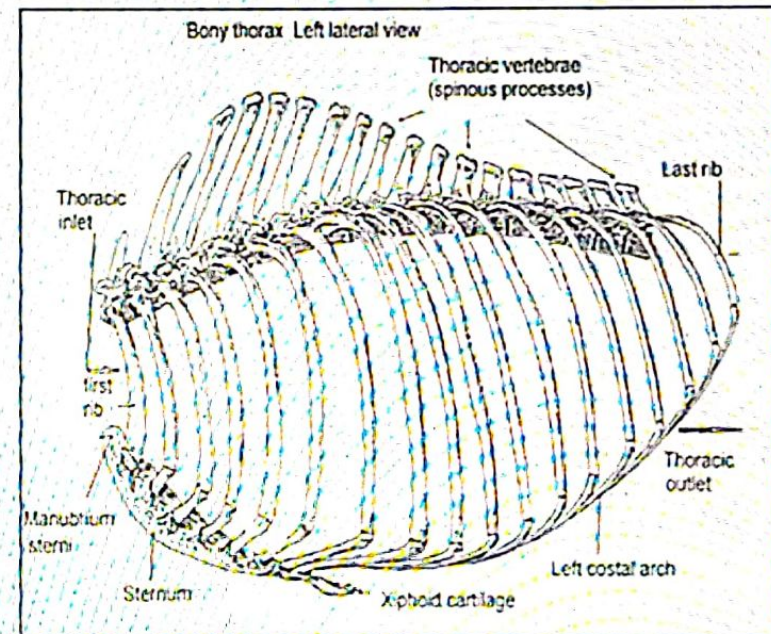
- The bodies of the thoracic vertebrae
- Intervertebral disc
- The ventral longitudinal ligament
- Longus colli muscle.

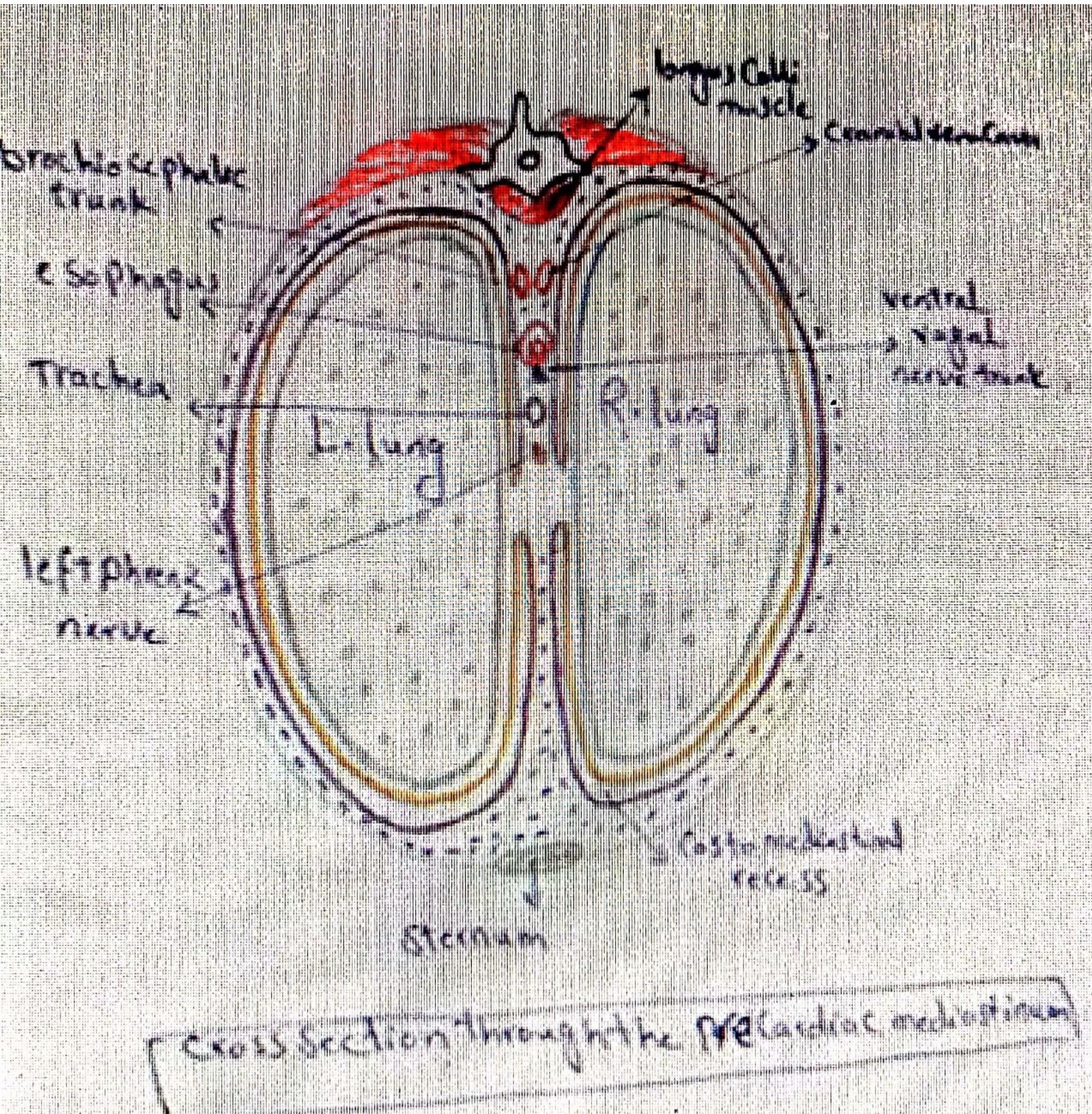
□ Lateral boundary

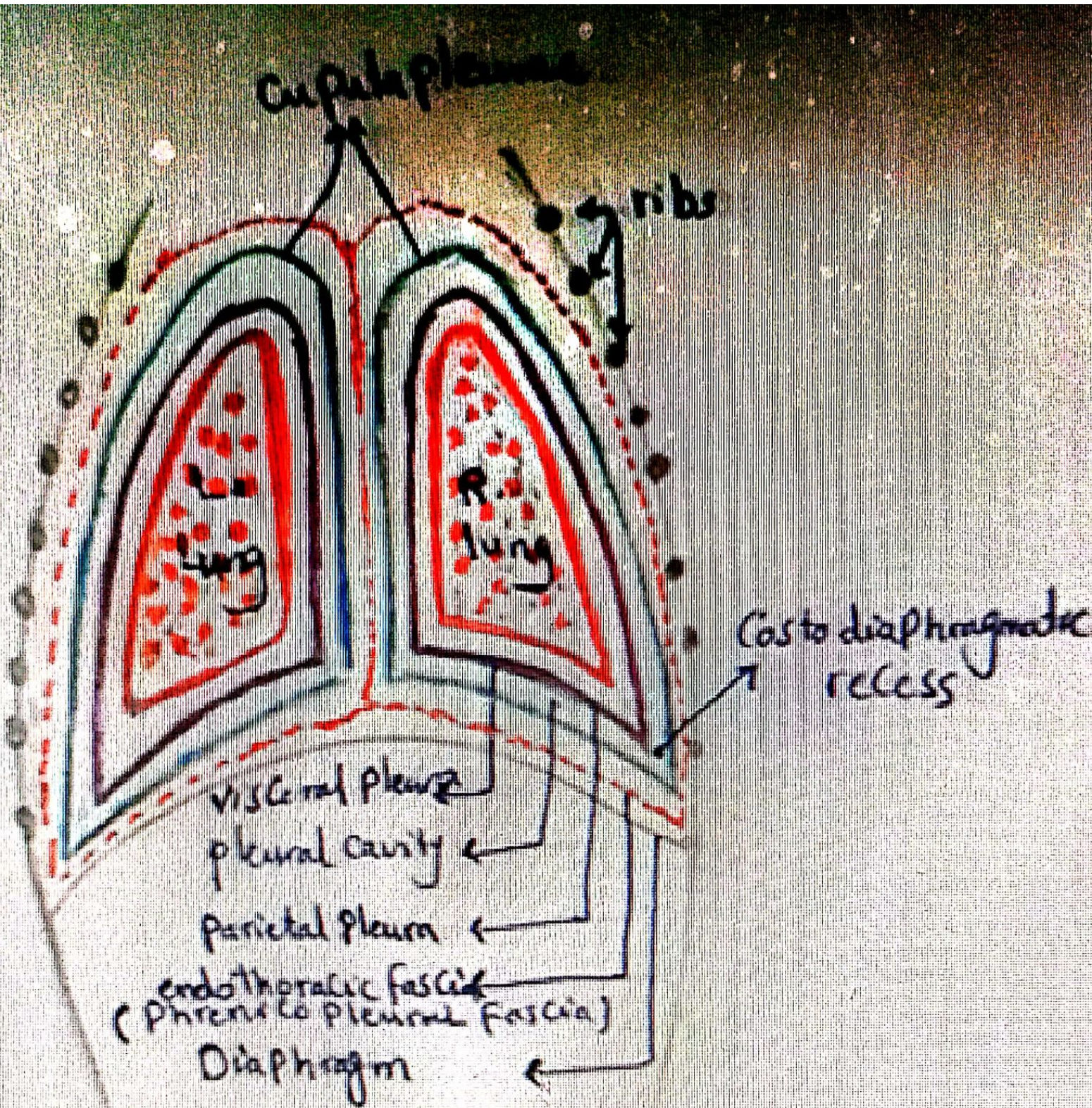
- The shafts of the ribs
- Costal cartilage
- The intercostal muscles.

□ Ventral boundary

- The sternum
- Transverse thoracic muscle

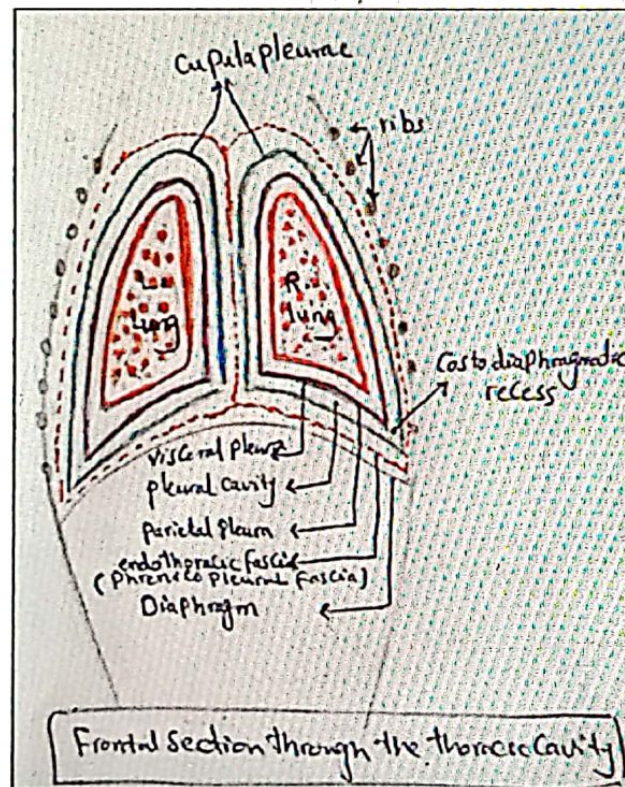






Frontal Section through the Thoracic Cavity

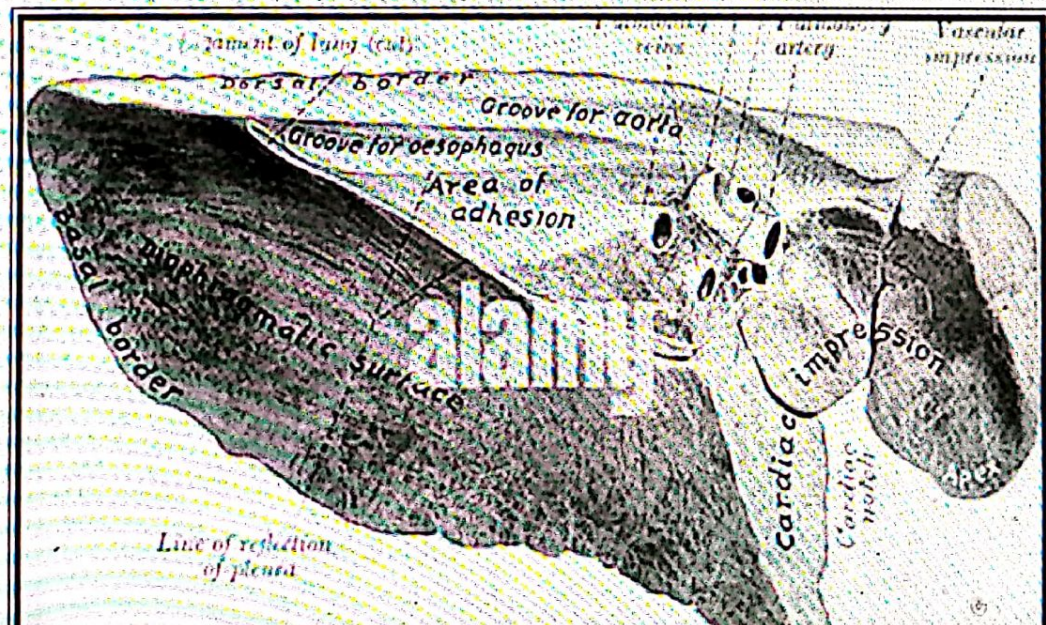
- At the median plane, it is reflected dorsally and ventrally to the thoracic cavity (**mediastinal fascia**) and divided it into two halves to divide the thoracic cavity into two chambers, each chamber is lined with its own serous membrane (**pleura**), which form the pleural sac enclosing its pleural cavity
- **The pleural cavity** is the area between visceral and parietal pleura



☐ The mediastinum is divided into 3 parts:

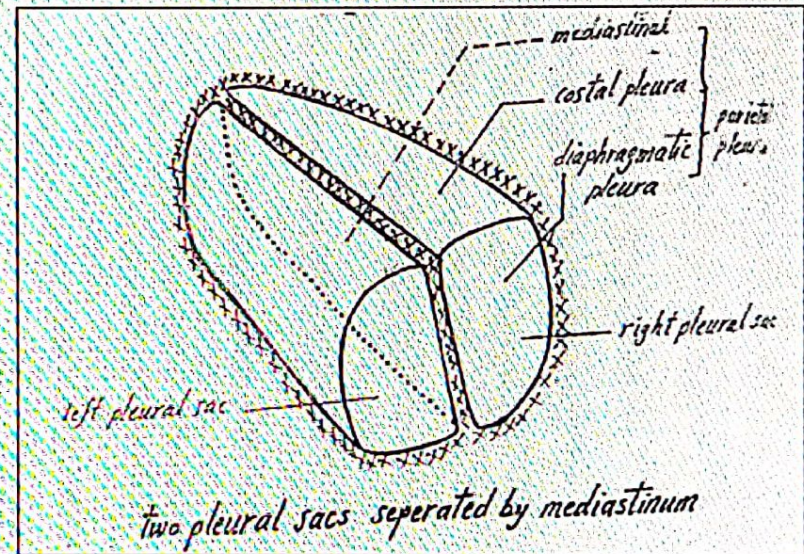
1. Cranial mediastinum = precardiac mediastinum
2. Middle mediastinum = cardiac mediastinum
3. Caudal mediastinum = post cardiac mediastinum

☐ **In the frontal section** that passes through the root of the lung, it may be divided into **dorsal and ventral mediastinum**

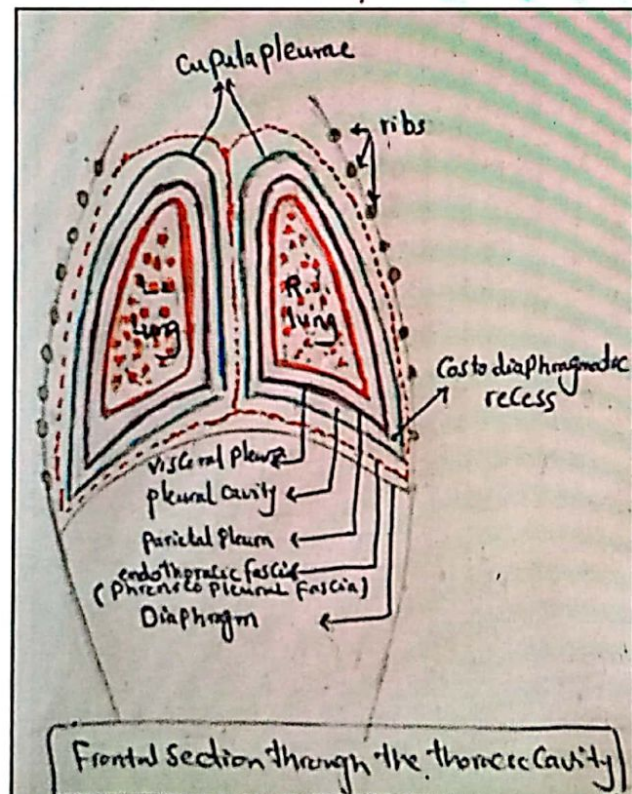


Mediastinum

- It is an interval between the two pleural sacs, extends from thoracic inlet until the diaphragm.
- It is bounded
 - dorsally by the vertebral column
 - laterally by the pleural sac
 - ventrally by the sternum
- All structures lies in thoracic cavity present at the mediastinum except the lung, the right phrenic nerve & the caudal vena cava.



- At the median plane, it is reflected dorsally and ventrally to the thoracic cavity (**mediastinal fascia**) and divided it into two halves to divide the thoracic cavity into two chambers, each chamber is lined with its own serous membrane (**pleura**), which form the pleural sac enclosing its pleural cavity
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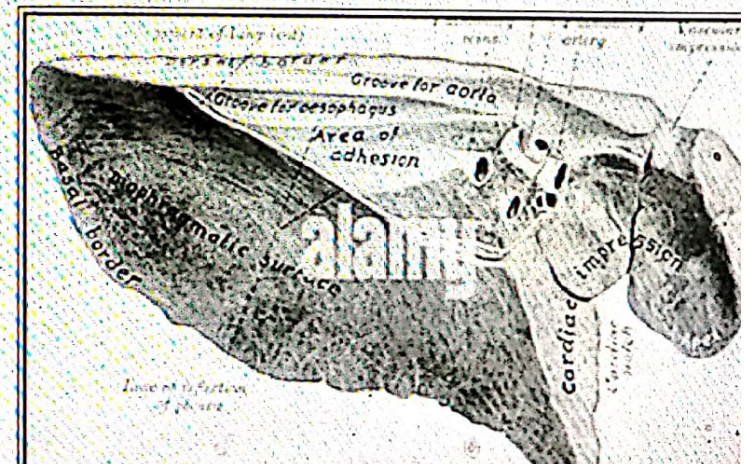
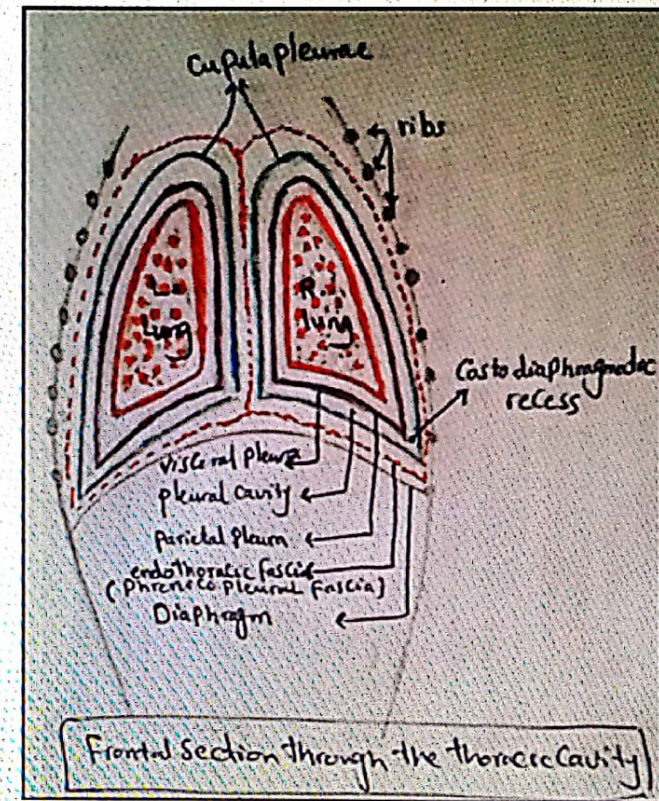


- The parietal pleura is subdivided according to the region into:

1. Costal pleura on the ribs.
2. Diaphragmatic pleura on diaphragm.
3. Mediastinal pleura at mediastinum.

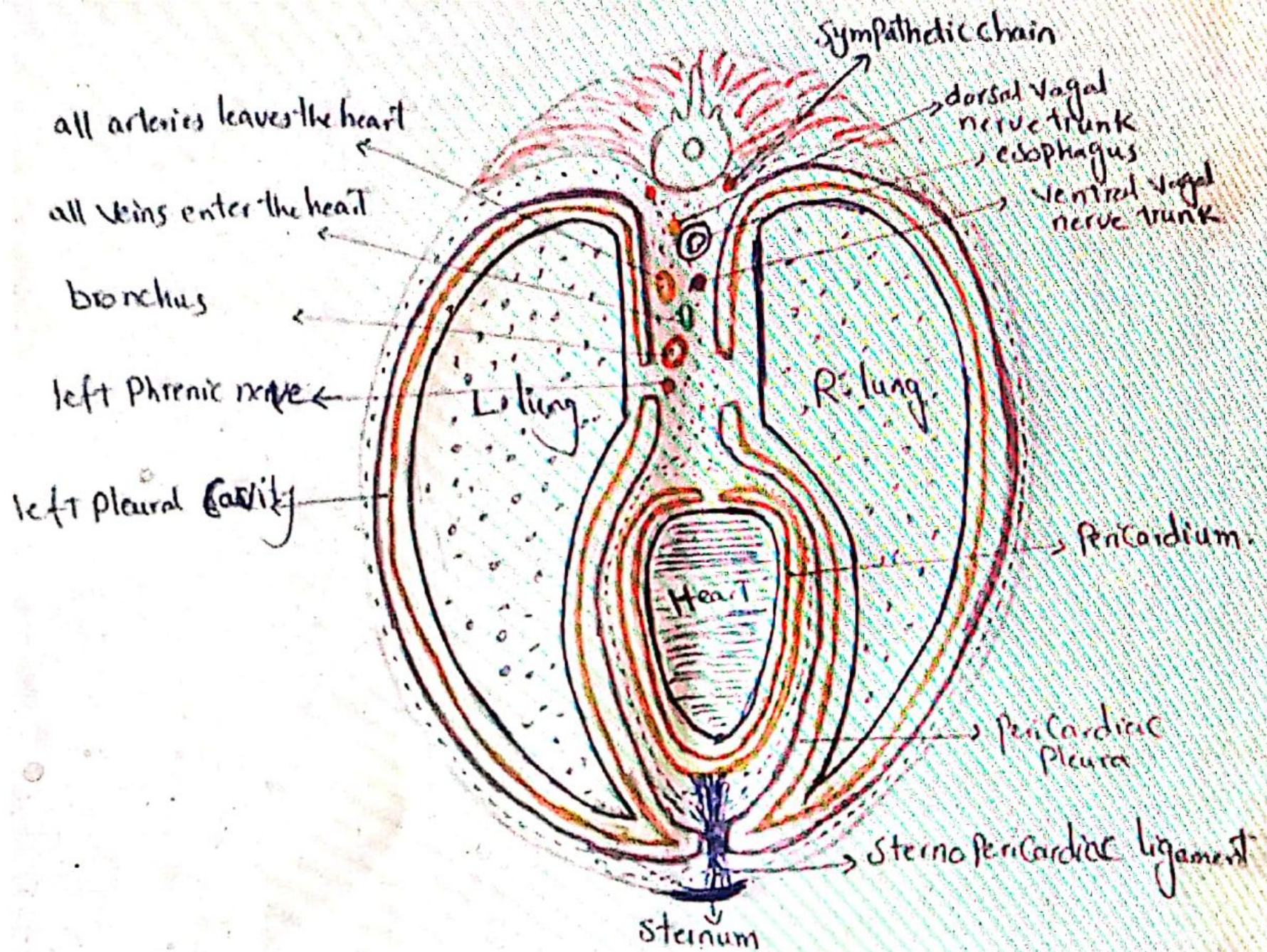
❑ **Pleural cavity** is the area between visceral and parietal pleura.

❑ **Pulmonary ligament** It isn't a true ligament but it is a reflection of the mediastinal pleura onto the lung. The reflection extends caudal to the root of the lung in a tapering fashion



Pleura

- It is a thin glistening, slippery serous membrane arranged in two sacs, area in between is called the mediastinum space .
- It is consisted of :
 1. Pulmonary or visceral pleura which cover the lung.
 2. Parietal pleura which forming the rest of pleural sac.



Cross section Through the Cardiac mediastinum

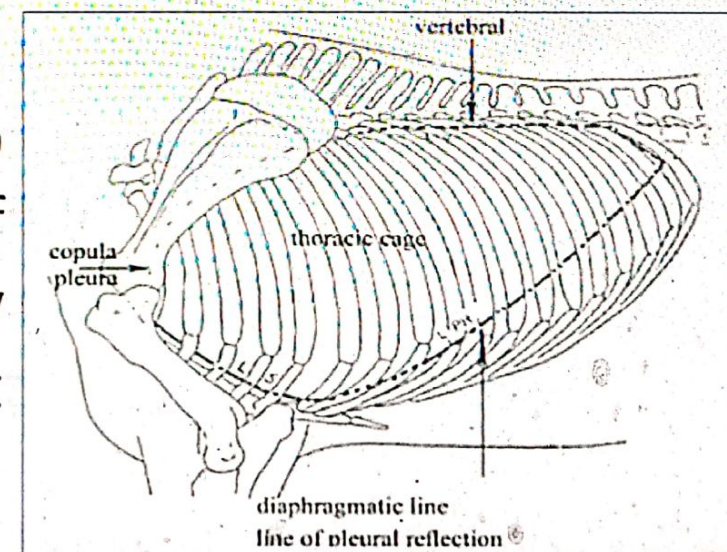
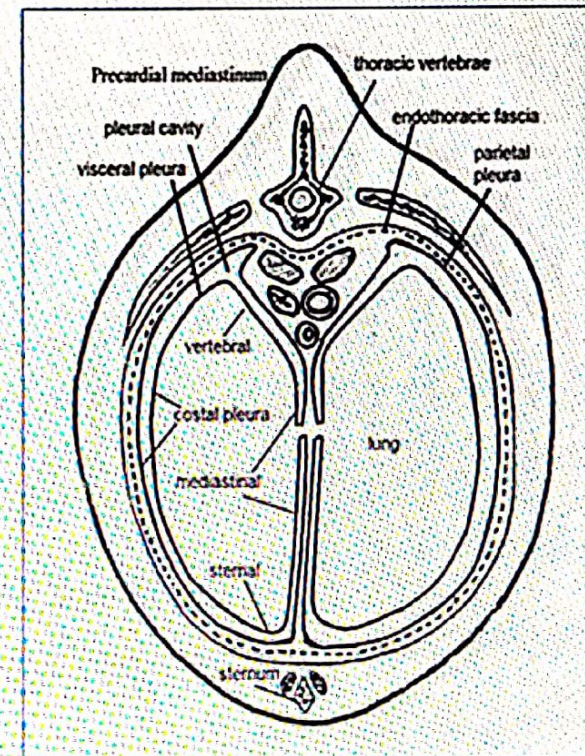
Lines of pleural reflection

- The costal pleura are reflected along three lines

1. **Vertebral line** ---- the costal pleura turns ventrally to form mediastinal pleura.

2. **Sternal line** ----- the costal pleura are reflected dorsally to become mediastinal pleura.

3. **Diaphragmatic line** ---- the costal pleura passes from the lateral wall to the diaphragm. From the sternal end of the 9th rib, curves gently, turns medially to ends at the vertebral end of the last intercostal space



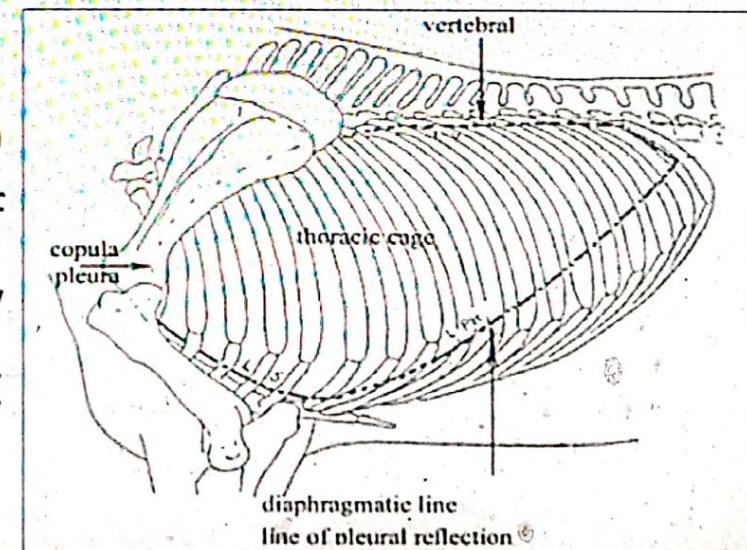
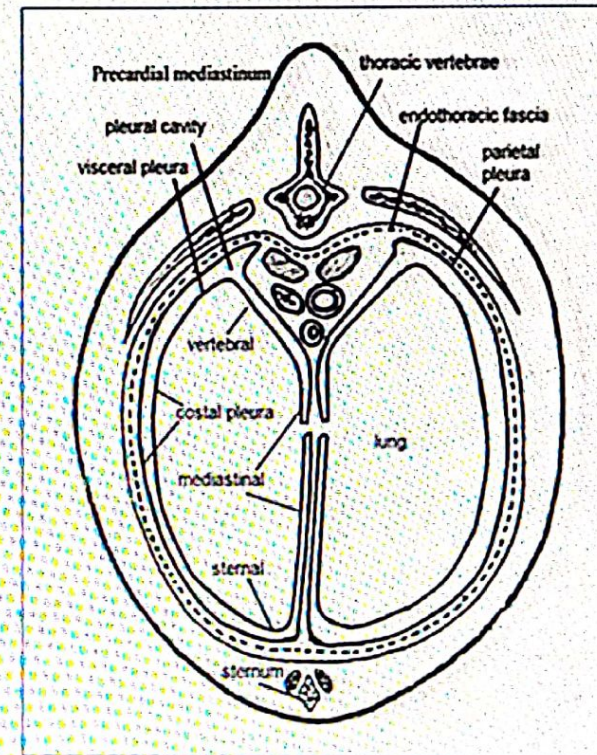
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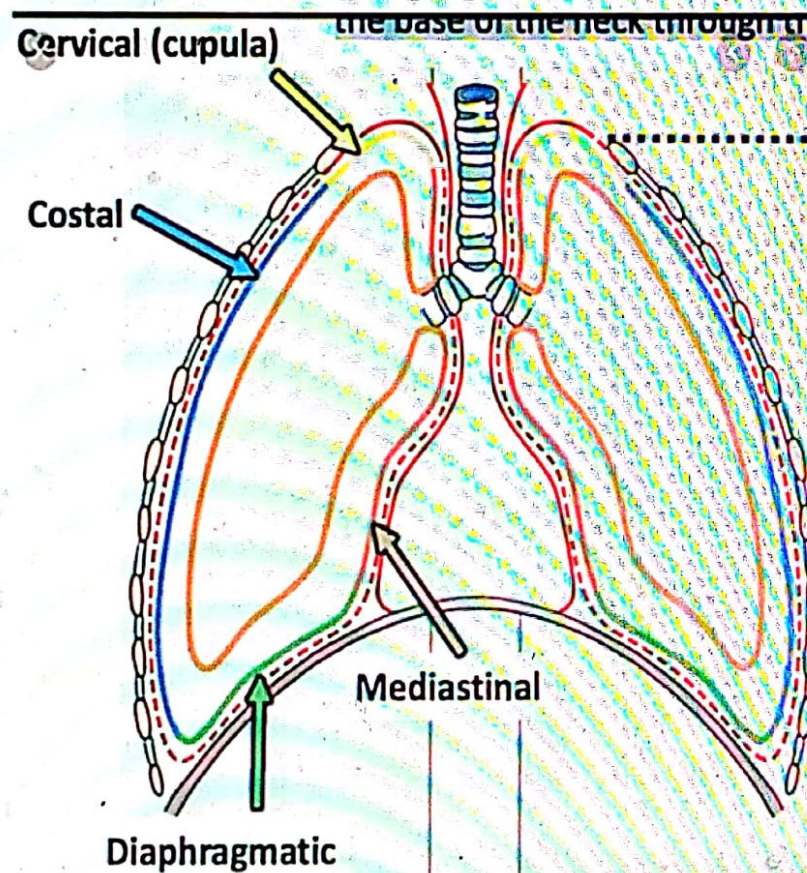


Thoracic cavity contains the following:

1. Cupula pleura
2. plica vena cava
3. Mediastinal recess
4. Costodiaphragmatic recesses
5. Costomediastinal recess

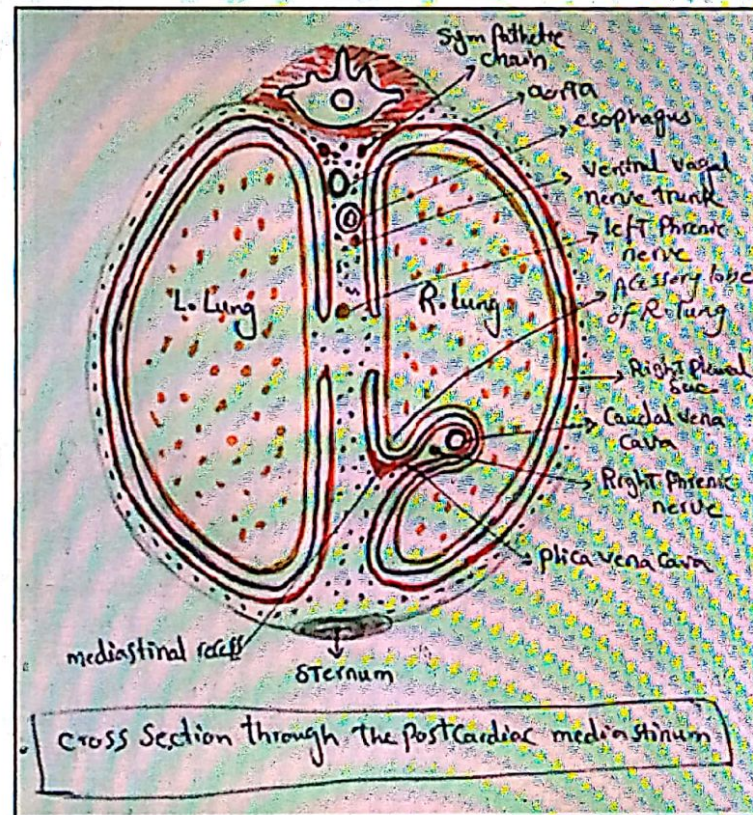
Cupula pleura

The parietal pleura forming a blind sac extend beyond the thoracic inlet into cervical region



Mediastinal recess

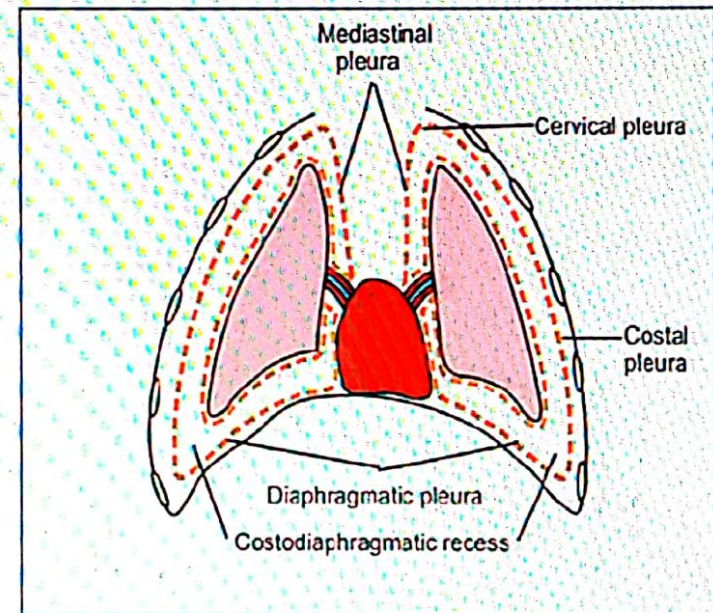
It is a part of right pleural cavity situated between the mediastinum and the plica vena cava. It contains the accessory lobe of the right lung.



Costodiaphragmatic recesses

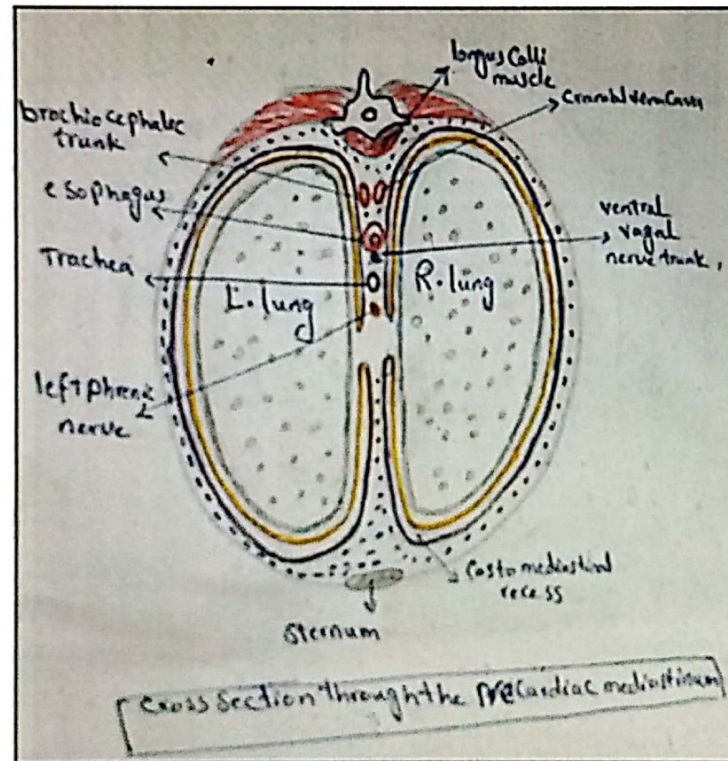
It is a space located at the junction of costal and diaphragmatic pleurae in each pleural cavity

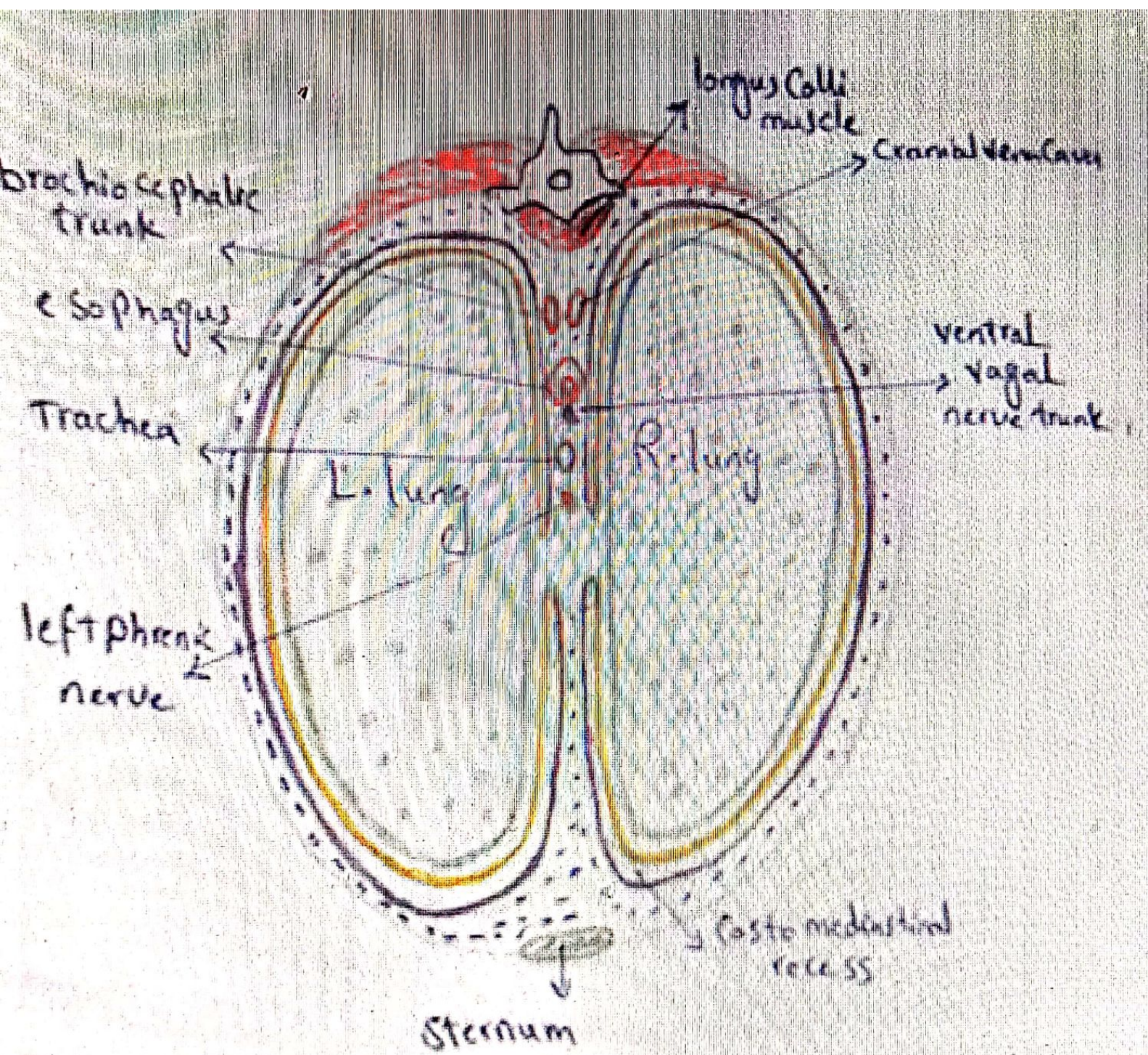
Formed when the diaphragmatic line of pleural reflection reflected from lateral thoracic wall on diaphragm



Costomediastinal recess

- It is a narrow angular recess formed by the sternal pleural reflection between the costal and mediastinal pleura





Cross Section through the precardiac mediastinum